

JAY KUO

Data & Automation Engineer · Production Control

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CORE SKILLS

Programming & Query	Python · SQL · R · Excel VBA
Automation & Reporting	VBA Automation · HTML Dashboards · Multi-DB Integration · BI Tools
Analytics & ML	Data Visualization · Statistics · Machine Learning (KMeans, Naïve Bayes)
Planning & Control	Production Control · Output Plan · Lot Planning · Fab Daily Status
Supply Chain & Process	Supply Chain Coordination · Process Optimization · Vendor Management
Collaboration	Cross-Functional Collaboration · Stakeholder Communication · Data-Driven Decisions
Languages	Mandarin (Native) · English (Fluent — TOEFL 107) · German (Introductory)

EXPERIENCE

TSMC

Hsinchu, Taiwan

Production Control Integration Planner

Jul 2024 – Present

- Engineered a **VBA-driven Fab Daily Status pipeline** that integrates data across multiple databases and refreshes daily, enabling multi-dimensional analysis (Customer view, Fab view, and more).
- Built an Excel VBA tool that **auto-queries SQL** to generate the weekly backend Output Plan, cutting ~6 hours of manual effort per week while improving accuracy.
- Created a team-accessible **"Pull-In Ratio" HTML report**, giving every user a self-serve, browser-based view of pull-in performance.
- Deployed a suite of **VBA automation tools** across team workflows, substantially raising work efficiency and standardizing reporting.
- Led verification-code development for the Early Wafer Start Candidate project, surfacing key issues projected to increase the candidate pool **2.4x**.
- Drove the **Lot Plan revamp** with IT, FPC, and APC teams to rebuild the CR3 modification tool.

Production Control Integration Planning Intern

Jul 2023 – Sep 2023

- Applied **KMeans clustering** for product-pattern analysis, improving ASRS picking efficiency; represented the department in the final intern competition.

Taiwan Institute of Economic Research

Taipei, Taiwan

Data Analyst

Sep 2021 – Aug 2022

- Aggregated **1,000+ biweekly data points** via SQL and interactive dashboards, boosting team productivity by 25%.
- Applied **Naïve Bayes sentiment analysis** across 6 large datasets and led quantitative-analysis training for 12+ researchers, supporting a renowned monthly publication.
- Evaluated economic data for Taiwan's SME market to support growth forecasting, reducing researcher workload by 20%.

Merck & Co., Inc.

Taipei, Taiwan

Global Procurement Intern

Sep 2019 – Aug 2020

- Built a self-service vendor-purchasing approval workflow, making orders **70% more efficient** and 25% faster to complete.
- Redesigned 6 procurement websites with UX improvements (~30% team workload) and cut operational expenses by 15% through data-driven process changes.
- Mentored 3 interns in Python, JavaScript, web design, business analysis, and optimization.

EDUCATION

Georgia Institute of Technology — MSc, Computer Science

Aug 2023 – Jul 2025

Specialization: Human-Computer Interaction. Coursework: Machine Learning for Trading, Knowledge-Based AI.

University of Washington — MSc, Information Management

Sep 2022 – Jun 2024

Specializations: Data Science, Business Intelligence, Cybersecurity. iSchool Merit-Based Scholarship.

National Chengchi University — BSc, International Business

Sep 2016 – Jun 2020

School Representative, Asian Economic Seminar (GPAC). English-taught program (Statistics, Economics).